Advanced Time Series Analysis PSET 1

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1 Question 1

1.1 (i)

We have that the $n \times n$ lag polynomial $C(L) = \sum_{\ell=0}^{\infty} C_\ell L^\ell$ satisfying the one-summability condition $\sum_{\ell=0}^{\infty} \ell |C_{ij,\ell}| < \infty$ for all i, j, and the lag polynomial $B(L) = \sum_{\ell=0}^{\infty} B_\ell L^\ell$ defined by

$$B_\ell = -\sum_{m=\ell+1}^{\infty} C_m, \qquad \ell = 0, 1, 2, \dots$$

We want to show that $(I_n - L)\, B(L) = C(L) - C(1)$.

We compute $(I_n - L)\, B(L) = B(L) - L\, B(L)$ by collecting powers of L:

$$\begin{aligned}(I_n - L)\, B(L) &= \sum_{\ell=0}^{\infty} B_\ell\, L^\ell \; - \; \sum_{\ell=0}^{\infty} B_\ell\, L^{\ell+1} = \sum_{\ell=0}^{\infty} B_\ell\, L^\ell \; - \; \sum_{\ell=1}^{\infty} B_{\ell-1}\, L^\ell \\ &= B_0 + \sum_{\ell=1}^{\infty} \big(B_\ell - B_{\ell-1}\big)\, L^\ell.\end{aligned}$$

For $\ell \geq 1$, the coefficient differences are:

$$B_\ell - B_{\ell-1} = -\sum_{m=\ell+1}^{\infty} C_m - \Big(-\sum_{m=\ell}^{\infty} C_m\Big) = \sum_{m=\ell}^{\infty} C_m - \sum_{m=\ell+1}^{\infty} C_m = C_\ell.$$

For the constant term:

$$B_0 = -\sum_{m=1}^{\infty} C_m = C_0 - \sum_{m=0}^{\infty} C_m = C_0 - C(1).$$

Substituting back:

$$(I_n - L)\, B(L) = C_0 - C(1) + \sum_{\ell=1}^{\infty} C_\ell\, L^\ell = \sum_{\ell=0}^{\infty} C_\ell\, L^\ell - C(1) = C(L) - C(1).$$

*Claude Code was used to assist with writing and debugging the Python code for Questions 3 and 4, and double checking my work and cleanliness of latex and presentation.

1.2 (ii)

We have $y_t = C(L)\,u_t$ with $u_t \sim \mathrm{WN}(0,\Sigma)$ and $C(0) = I_n$. We want to show there exists a covariance stationary process $\{\eta_t\}$ such that $y_t = C(1)\,u_t + \eta_t - \eta_{t-1}$. From part (i), we have $(I_n - L)\,B(L) = C(L) - C(1)$, i.e.,

$$C(L) = C(1) + (1 - L)\,B(L).$$

Let $\eta_t \equiv B(L)\,u_t = \sum_{\ell=0}^{\infty} B_\ell\,u_{t-\ell}$. Since $B(L)$ is absolutely summable, η_t is a well-defined, covariance stationary VMA(∞) process. Applying $C(L)$ to u_t we get:

$$y_t = C(L)\,u_t = C(1)\,u_t + (1 - L)\,B(L)\,u_t = C(1)\,u_t + \eta_t - \eta_{t-1}.$$

2 Question 2

We know $\{y_t\}$ and $\{x_t\}$ are scalar, jointly covariance stationary, mean-zero time series, with $y_t = C(L)\,u_t$, $C(0) = 1$, $u_t \sim \mathrm{WN}(0,\sigma_u^2)$, and $u_t \in \mathrm{sp}(y_\tau,\, -\infty < \tau \le t)$. Since $y_t = C(L)\,u_t$ implies $\mathrm{sp}(y_\tau,\, \tau \le t) \subseteq \mathrm{sp}(u_\tau,\, \tau \le t)$, and $u_t \in \mathrm{sp}(y_\tau,\, \tau \le t)$ gives the reverse inclusion, we have

$$\mathrm{sp}(y_\tau,\, -\infty < \tau \le t) = \mathrm{sp}(u_\tau,\, -\infty < \tau \le t).$$

2.1 (i)

We want to show that $\mathrm{Var}^*(y_{t+1} \mid \{y_\tau\}_{-\infty<\tau\le t}) = \sigma_u^2$.

By the span equality above, $(y_\tau,\, \tau \le t) = (u_\tau,\, \tau \le t)$. We then get:

$$y_{t+1} = \sum_{\ell=0}^{\infty} C_\ell\,u_{t+1-\ell} = C_0\,u_{t+1} + \sum_{\ell=1}^{\infty} C_\ell\,u_{t+1-\ell}.$$

The second sum lies in $(u_\tau,\, \tau \le t)$, and since u_t is white noise, $E^*(u_{t+1} \mid \{u_\tau\}_{\tau\le t}) = 0$. Therefore:

$$E^*(y_{t+1} \mid \{y_\tau\}_{\tau\le t}) = \sum_{\ell=1}^{\infty} C_\ell\,u_{t+1-\ell}.$$

The forecast error is $y_{t+1} - E^*(y_{t+1} \mid \{y_\tau\}_{\tau\le t}) = C_0\,u_{t+1} = u_{t+1}$ (since $C_0 = 1$), so

$$\mathrm{Var}^*(y_{t+1} \mid \{y_\tau\}_{\tau\le t}) = \mathbb{E}[u_{t+1}^2] = \sigma_u^2.$$

2.2 (ii)

We want to show that $\mathrm{Var}^*(y_{t+1} \mid \{y_\tau, x_\tau\}_{\tau\le t}) = \mathrm{Var}^*(u_{t+1} \mid \{u_\tau, x_\tau\}_{\tau\le t})$.

By the span equality, $(y_\tau,\, \tau \le t) = (u_\tau,\, \tau \le t)$, so $(y_\tau, x_\tau;\, \tau \le t) = (u_\tau, x_\tau;\, \tau \le t)$.

Then

$$y_{t+1} = u_{t+1} + \sum_{\ell=1}^{\infty} C_\ell\,u_{t+1-\ell},$$

where $\sum_{\ell=1}^{\infty} C_\ell\,u_{t+1-\ell} \in (u_\tau,\, \tau \le t) \subseteq (u_\tau, x_\tau;\, \tau \le t)$. Therefore:

$$E^*\big(y_{t+1} \mid \{u_\tau, x_\tau\}_{\tau\le t}\big) = E^*\big(u_{t+1} \mid \{u_\tau, x_\tau\}_{\tau\le t}\big) + \sum_{\ell=1}^{\infty} C_\ell\,u_{t+1-\ell},$$

so the forecast error is:

$$y_{t+1} - E^*(y_{t+1} \mid \cdots) = u_{t+1} - E^*(u_{t+1} \mid \{u_\tau, x_\tau\}_{\tau \le t}).$$

Taking the variance of both sides and using $(y_\tau, x_\tau;\, \tau \le t) = (u_\tau, x_\tau;\, \tau \le t)$ gives $\mathrm{Var}^*(y_{t+1} \mid \{y_\tau, x_\tau\}_{\tau \le t}) = \mathrm{Var}^*(u_{t+1} \mid \{u_\tau, x_\tau\}_{\tau \le t})$.

2.3 (iii)

We want to show that $\sigma_u^2 = \mathrm{Var}^*(u_{t+1} \mid \{u_\tau, x_\tau\}_{\tau \le t})$ if and only if $\mathrm{Cov}(u_{t+1}, x_{t-\ell}) = 0$ for all $\ell \ge 0$.

Since $u_t \sim \mathrm{WN}(0, \sigma_u^2)$, we have $u_{t+1} \perp (u_\tau,\, \tau \le t)$ (orthogonal in L^2). Therefore:

$$E^*(u_{t+1} \mid \{u_\tau, x_\tau\}_{\tau \le t}) = 0 \quad \iff \quad u_{t+1} \perp (u_\tau, x_\tau;\, \tau \le t).$$

Since u_{t+1} is already orthogonal to all u_τ $(\tau \le t)$, the orthogonality to the entire joint span holds if and only if $u_{t+1} \perp x_{t-\ell}$ for all $\ell \ge 0$, i.e., $\mathrm{Cov}(u_{t+1}, x_{t-\ell}) = 0$ for all $\ell \ge 0$.

When this holds, $E^*(u_{t+1} \mid \cdots) = 0$ and $\mathrm{Var}^*(u_{t+1} \mid \cdots) = \mathbb{E}[u_{t+1}^2] = \sigma_u^2$. Conversely, if $\mathrm{Var}^* = \sigma_u^2$, then the projection equals zero (since $\mathrm{Var}^* = \sigma_u^2 - \mathbb{E}[(E^*(\cdots))^2]$), which implies the orthogonality conditions.

2.4 (iv)

We want to show that for any $\ell \in \mathbb{Z} \cup \{\infty\}$,

$$E^*\big(x_t \mid \{y_\tau\}_{-\infty < \tau \le t+\ell}\big) = \sum_{m=-\infty}^{\ell} a_m\, u_{t+m}, \qquad a_m = \frac{\mathrm{Cov}(x_t,\, u_{t+m})}{\sigma_u^2}.$$

By the span equality, $(y_\tau,\, \tau \le t+\ell) = (u_\tau,\, \tau \le t+\ell)$. The best linear predictor of x_t given $\{u_\tau\}_{\tau \le t+\ell}$ is a linear combination $\hat{x}_t = \sum_{m=-\infty}^{\ell} a_m\, u_{t+m}$.

By orthogonality, the coefficients a_m must satisfy:

$$\mathbb{E}\big[(x_t - \hat{x}_t)\, u_{t+k}\big] = 0 \quad \text{for all } k \le \ell.$$

Which gives:

$$\mathbb{E}[x_t\, u_{t+k}] - \sum_{m=-\infty}^{\ell} a_m\, \mathbb{E}[u_{t+m}\, u_{t+k}] = 0.$$

Since u is white noise, $\mathbb{E}[u_{t+m}\, u_{t+k}] = \sigma_u^2\, \mathbf{1}(m = k)$. Thus we conclude that:

$$\mathrm{Cov}(x_t,\, u_{t+k}) - a_k\, \sigma_u^2 = 0 \quad \Longrightarrow \quad a_k = \frac{\mathrm{Cov}(x_t,\, u_{t+k})}{\sigma_u^2}.$$

2.5 (v)

We want to show that:

$$\mathrm{Var}^*(y_{t+1} \mid \{y_\tau, x_\tau\}_{\tau \le t}) = \mathrm{Var}^*(y_{t+1} \mid \{y_\tau\}_{\tau \le t})$$

if and only if

$$\mathrm{Var}^*(x_t \mid \{y_\tau\}_{-\infty < \tau < \infty}) = \mathrm{Var}^*(x_t \mid \{y_\tau\}_{\tau \le t}).$$

By (i) and (ii):

$$\text{LHS} \iff \text{Var}^*(u_{t+1} \mid \{u_\tau, x_\tau\}_{\tau \le t}) = \sigma_u^2.$$

By (iii), this holds if and only if $\text{Cov}(u_{t+1}, x_{t-\ell}) = 0$ for all $\ell \ge 0$. By stationarity, this is equivalent to

$$a_m = \frac{\text{Cov}(x_t, u_{t+m})}{\sigma_u^2} = 0 \quad \text{for all } m \ge 1. \tag{1}$$

By (iv) with $\ell = 0$:

$$E^*(x_t \mid \{y_\tau\}_{\tau \le t}) = \sum_{m=-\infty}^{0} a_m \, u_{t+m}.$$

By (iv) with $\ell = \infty$:

$$E^*(x_t \mid \{y_\tau\}_{-\infty < \tau < \infty}) = \sum_{m=-\infty}^{\infty} a_m \, u_{t+m}.$$

Therefore:

$$\begin{aligned}\text{Var}^*(x_t \mid \{y_\tau\}_{\tau \le t}) - \text{Var}^*(x_t \mid \{y_\tau\}_{-\infty<\tau<\infty}) &= \mathbb{E}\big[(E^*(x_t \mid \text{all}))^2\big] - \mathbb{E}\big[(E^*(x_t \mid \text{past}))^2\big] \\ &= \sum_{m=1}^{\infty} a_m^2 \, \sigma_u^2.\end{aligned}$$

So $\text{Var}^*(x_t \mid \{y_\tau\}_{-\infty<\tau<\infty}) = \text{Var}^*(x_t \mid \{y_\tau\}_{\tau \le t})$ if and only if $a_m = 0$ for all $m \ge 1$.

Both sides reduce to the same condition ($a_m = 0$ for all $m \ge 1$) which completes the proof.

3 Question 3

3.1 (i)

We transform the Federal Funds Rate to quarterly averages. Real GDP and the GDP deflator are transformed to annualized quarter-over-quarter log growth rates: $400 \times \Delta \log(X_t)$.

The three VAR variables are:

1. Real GDP growth: $y_{1,t} = 400\, \Delta \log(\text{GDPC1}_t)$,
2. GDP deflator inflation: $y_{2,t} = 400\, \Delta \log(\text{GDPDEF}_t)$,
3. Real rate spread: $y_{3,t} = \text{FEDFUNDS}_t - y_{2,t}$.

We fix the effective sample size across all lag orders by setting aside the first $p_{\max} = 20$ observations as the conditioning set. This gives a fixed effective sample of $T = 242$ for every p. We compute the BIC and AIC, which give:

$$\text{BIC selects } p = 1. \qquad \text{AIC selects } p = 11.$$

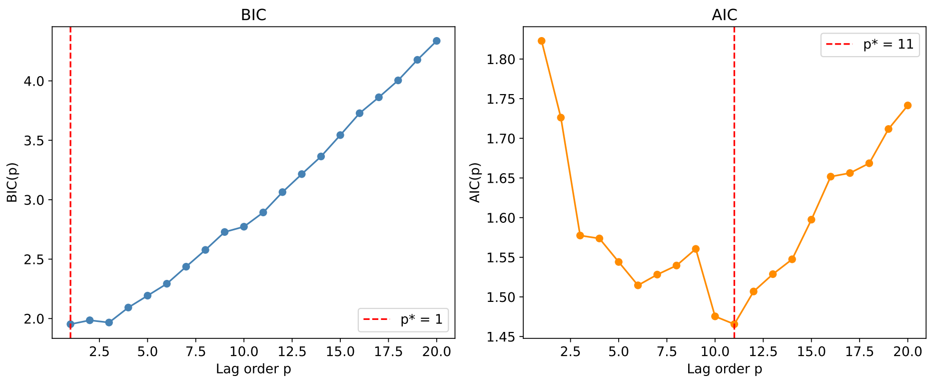

Figure 1: BIC and AIC as functions of VAR lag order p.

4 Question 4

We use a conjugate NIW prior:

$$\Sigma \sim \mathrm{IW}(\Psi, d), \qquad (\beta \mid \Sigma) \sim N(b, \Sigma \otimes \Omega),$$

with $d = n + 2 = 5$, $\Psi = (d - n - 1) \cdot 0.02^2\, I_n = 0.0004\, I_3$, $b = 0$. We now derive the diagonal matrix Ω.

We know that $\mathrm{Var}(C \mid \Sigma) = 10^2\, \Sigma$, which gives $\Omega_{1,1} = 100$. For the lag coefficients:

$$\mathrm{Cov}(B_{ij,s},\, B_{hm,r} \mid \Sigma) = \frac{0.2^2}{0.02^2/(d-n-1)}\, \frac{1}{s^2}\, \Sigma_{ih} = \frac{100}{s^2}\, \Sigma_{ih}, \qquad m = j,\ r = s.$$

In the Kronecker structure $\mathrm{Cov}(\beta \mid \Sigma) = \Sigma \otimes \Omega$, the element $\mathrm{Cov}(B_{ij,s},\, B_{hm,r} \mid \Sigma) = \Sigma_{ih} \cdot \Omega_{(s-1)n+1+j,\,(r-1)n+1+m}$, so we need

$$\Omega_{(s-1)n+1+j,\,(s-1)n+1+j} = \frac{100}{s^2}, \qquad s = 1, \dots, p,\ j = 1, \dots, n.$$

4.1 (i)

We condition on the first $p_{\max} = 20$ observations and use the resulting fixed effective sample $T = T_{\text{full}} - p_{\max} = 242$ for all lag orders, so that the marginal likelihoods are directly comparable. We compute the log marginal likelihood for each $p = 0, 1, \dots, 20$ using the expression in the GLP notes:

$$\begin{aligned}\log p(Y \mid \gamma) = &-\frac{nT}{2}\log(\pi) + \log\Gamma_n\Big(\frac{T+d}{2}\Big) - \log\Gamma_n\Big(\frac{d}{2}\Big) - \frac{T}{2}\log|\Psi| \\ &- \frac{n}{2}\log\big|D_\Omega'\, x'x\, D_\Omega + I_k\big| - \frac{T+d}{2}\log\big|I_n + D_\Psi'\, S\, D_\Psi\big|,\end{aligned}$$

where T denotes the effective sample size, $D_\Omega D_\Omega' = \Omega$, $D_\Psi D_\Psi' = \Psi^{-1}$, $\hat{B} = (x'x + \Omega^{-1})^{-1}x'y$, and $S = \hat{\varepsilon}'\hat{\varepsilon} + \hat{B}'\Omega^{-1}\hat{B}$.

The log marginal likelihood peaks at $p^* = 1$ (Figure 2) and decreases for $p > 1$, so we have a clear optimal $p^* = 1$ to choose.

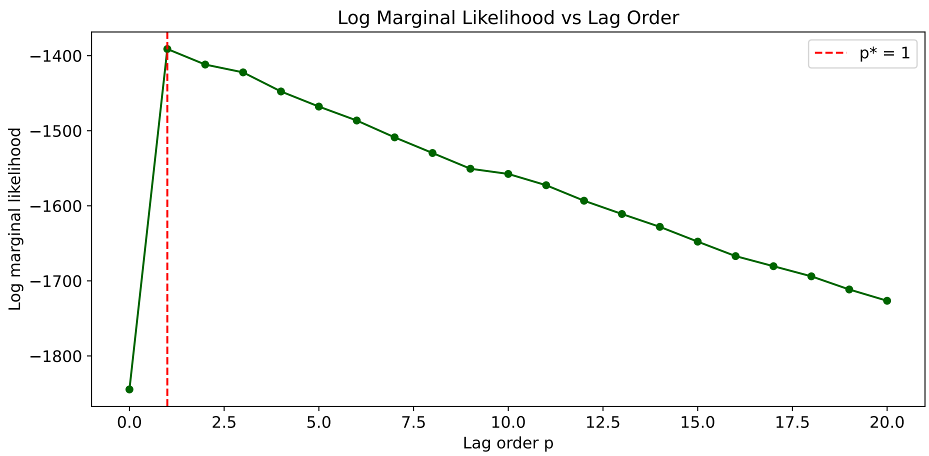

Figure 2: Log marginal likelihood as a function of the VAR lag order p.

4.2 (ii)

Using $p^* = 1$, we draw from the posterior predictive distribution of real GDP growth in 2020-Q2. We do so using the following steps:

1. Draw $\Sigma^{(i)}$ from the posterior $\mathrm{IW}\big(\Psi + S,\ T + d\big)$.
2. Draw $B^{(i)} \mid \Sigma^{(i)}$ from $N\big(\mathrm{vec}(\hat{B}),\ \Sigma^{(i)} \otimes (x'x + \Omega^{-1})^{-1}\big)$.
3. Given $(B^{(i)}, \Sigma^{(i)})$ and the observed data, simulate $y_{T+1}^{(i)}$ and then $y_{T+2}^{(i)}$ from the VAR model.
4. Record $y_{1,T+2}^{(i)}$, the real GDP growth component.
5. Repeat steps 1–4 for $i = 1, \ldots, 20{,}000$ to approximate the marginal predictive density.

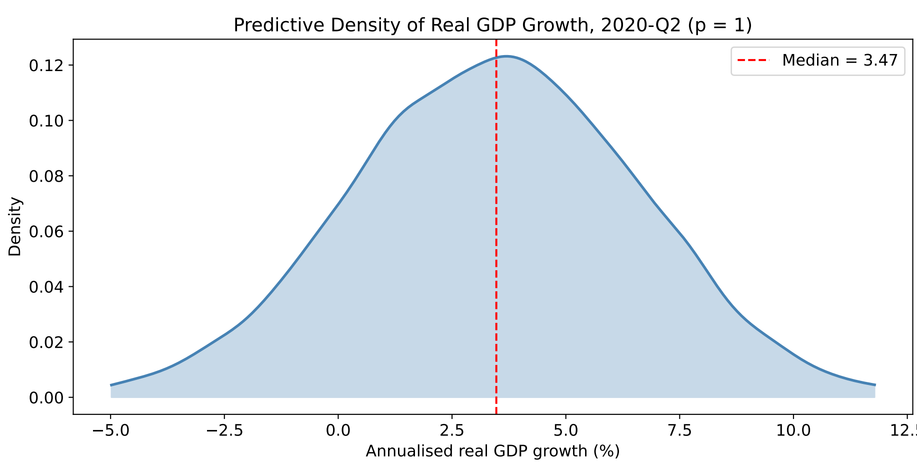

Figure 3: Posterior predictive density of annualized real GDP growth in 2020-Q2.

The posterior predictive distribution is approximately normal, centered around 3.4% annualized growth, with a standard deviation of about 3.2 percentage points. The 90% credible interval is roughly $[-1.9\%,\ 8.6\%]$ which is pretty far off from the actual due to the COVID shock.